BigQuerry Project on Public Google data analytics dataset

Table Schema on BigQuery: https://support.google.com/analytics/answer/3437719?hl=en

Query 01: calculate total visit, pageview, transaction and revenue for Jan, Feb and March 2017 order by month

```
SELECT
  FORMAT_DATE('%Y%m', parse_DATE('%Y%m%d',date)) AS month,
  SUM(totals.visits) AS visits,
  SUM(totals.pageviews) AS pageviews,
  SUM(totals.transactions) AS transactions,
  SUM(totals.transactionRevenue)/1000000 AS revenue,
FROM
  `bigquery-public-data.google_analytics_sample.ga_sessions_*`
WHERE
  _table_suffix BETWEEN '20170101'
  AND '20170331'
GROUP BY
  month
ORDER BY
  month;
```

-- Result Output:

Row	month	visits	pageviews	transactions	revenue
1	201701	64694	257708	713	97877.79
2	201702	62192	233373	733	108756.52
3	201703	69931	259522	993	130964.27

Query 02: Bounce rate per traffic source in July 2017

```
SELECT
  trafficSource.source,
  SUM(totals.visits) AS total_visits,
  SUM(totals.bounces) AS total_no_of_bounces,
  (SUM(totals.bounces)/SUM(totals.visits)*100) AS bounce_rate
FROM
  `bigquery-public-data.google_analytics_sample.ga_sessions_201707*`
WHERE
  _table_suffix BETWEEN '01'
  AND '31'
GROUP BY
  trafficSource.source
ORDER BY
  total_visits DESC;
```

-- Result Output:

Row	source	total_visits	total_no_of_bounces	bounce_rate
1	google	38400	19798	51.55729166666667
2	(direct)	19891	8606	43.265798602382986
3	youtube.com	6351	4238	66.72964887419303
4	analytics.google.com	1972	1064	53.9553752535497
5	Partners	1788	936	52.348993288590606

Query 3: Revenue by traffic source by week, by month in June 2017

```
SELECT
    'Week' as time_type,
    format_date('%Y%W', parse_date('%Y%m%d', date)) AS time,
    trafficSource.source,
    sum(totals.totalTransactionRevenue)/10000 AS revenue
FROM
  `bigquery-public-data.google_analytics_sample.ga_sessions_201706*`
Where _table_suffix between '01' and '30'
group by time, trafficSource.source
UNION ALL
SELECT
    'Month' as time_type,
    format_date('%Y%m', parse_date('%Y%m%d', date)) AS time,
    trafficSource.source,
    sum(totals.totalTransactionRevenue)/10000 AS revenue
FROM
  `bigquery-public-data.google_analytics_sample.ga_sessions_201706*`
Where _table_suffix between '01' and '31'
group by time, trafficSource.source
order by revenue desc;


with month_data as(
SELECT
  "Month" as time_type,
  format_date("%Y%m", parse_date("%Y%m%d", date)) as month,
  trafficSource.source AS source,
  SUM(totals.totalTransactionRevenue)/1000000 AS revenue
FROM `bigquery-public-data.google_analytics_sample.ga_sessions_*`
WHERE
_TABLE_SUFFIX BETWEEN '20170601' AND '20170631'
GROUP BY 1,2,3
order by revenue DESC
),

week_data as(
```

```
SELECT
  "Week" as time_type,
  format_date("%Y%W", parse_date("%Y%m%d", date)) as date,
  trafficSource.source AS source,
  SUM(totals.totalTransactionRevenue)/1000000 AS revenue
FROM `bigquery-public-data.google_analytics_sample.ga_sessions_*`
WHERE
_TABLE_SUFFIX BETWEEN '20170601' AND '20170631'
GROUP BY 1,2,3
order by revenue DESC
)

select * from month_data
union all
select * from week_data
```

--Result output:

Job information | Results | JSON | Execution details

Row	time_type	time	source	revenue
1	Month	201706	(direct)	9723162.0
2	Week	201724	(direct)	3088391.0
3	Week	201725	(direct)	2725432.0
4	Month	201706	google	1875718.0
5	Week	201723	(direct)	1730268.0

Query 04: Average number of product pageviews by purchaser type (purchasers vs non-purchasers) in June, July 2017.

-- Note: totals.transactions >=1 for purchaser and totals.transactions is null for non-purchaser

```
WITH
  purchaser_view AS (
  SELECT
    FORMAT_DATE('%Y%m', parse_DATE('%Y%m%d',
        date)) AS month,
    (SUM(totals.pageviews)/COUNT(DISTINCT fullVisitorId)) AS avg_pageviews_purchase,
  FROM
    `bigquery-public-data.google_analytics_sample.ga_sessions_2017*`
  WHERE
    _table_suffix BETWEEN '0601'
    AND '0731'
    AND totals.transactions IS NOT NULL
  GROUP BY
    month),
  non_purchaser_view AS (
  SELECT
```

```
    FORMAT_DATE('%Y%m', parse_DATE('%Y%m%d',
        date)) AS month,
    (SUM(totals.pageviews)/COUNT(DISTINCT fullVisitorId)) AS avg_pageviews_non_purchase,
  FROM
    `bigquery-public-data.google_analytics_sample.ga_sessions_2017*`
  WHERE
    _table_suffix BETWEEN '0601'
    AND '0731'
    AND totals.transactions IS NULL
  GROUP BY
    month)
SELECT
  purchaser_view.month,
  avg_pageviews_purchase,
  avg_pageviews_non_purchase
FROM
  purchaser_view
LEFT JOIN
  non_purchaser_view
ON
  purchaser_view.month=non_purchaser_view.month
ORDER BY
  month
```

-- Result output:

Row	month	avg_pageviews_purchase	avg_pageviews_non_purchase
1	201706	25.735763097949885	4.074559876184948
2	201707	27.720954356846473	4.191840874707743

Query 05: Average number of transactions per user that made a purchase in July 2017

```
SELECT
  FORMAT_DATE('%Y%m', parse_DATE('%Y%m%d',
      date)) AS month,
  SUM(totals.transactions)/COUNT(DISTINCT fullVisitorId) AS Avg_total_transactions_per_user
FROM
  `bigquery-public-data.google_analytics_sample.ga_sessions_201707*`
WHERE
  _table_suffix BETWEEN '01'
  AND '31'
  AND totals.transactions>=1
GROUP BY
  month
```

-- Result output:

Row	month	Avg_total_transactions_per_user
1	201707	1.112033195020747

Query 06: Average amount of money spent per session. Only include purchaser data in July 2017

```
SELECT
  FORMAT_DATE('%Y%m', parse_DATE('%Y%m%d',
      date)) AS month,
  SUM(totals.transactionRevenue)/SUM(totals.visits) AS avg_revenue_by_user_per_visit
FROM
  `bigquery-public-data.google_analytics_sample.ga_sessions_201707*`
WHERE
  _table_suffix BETWEEN '01'
  AND '31'
  AND totals.transactions IS NOT NULL
GROUP BY
  month
```

-- Result output:

Row	month	avg_revenue_by_user_per_visit
1	201707	1.2075584869059166E8

Query 07: Products purchased by customers who purchased product A (Classic Ecommerce)

```
SELECT
  product.v2ProductName AS other_pruchased_products,
  SUM(product.productQuantity) AS productQuantity
FROM
  `bigquery-public-data.google_analytics_sample.ga_sessions_201707*`,
  UNNEST(hits) AS hits,
  UNNEST(hits.product) AS product
WHERE
  _table_suffix BETWEEN '01'
  AND '31'
  AND fullVisitorId IN (
  SELECT
    fullVisitorId
  FROM
    `bigquery-public-data.google_analytics_sample.ga_sessions_201707*`,
    UNNEST(hits) AS hits,
    UNNEST(hits.product) AS product
  WHERE
    _table_suffix BETWEEN '01'
    AND '31'
    AND product.v2ProductName= "YouTube Men's Vintage Henley"
    AND product.productRevenue IS NOT NULL )
  AND product.productRevenue IS NOT NULL
  AND product.v2ProductName<>"YouTube Men's Vintage Henley"
GROUP BY
  other_pruchased_products
ORDER BY
  productQuantity DESC
```

--Result output:

Row	other_pruchased_products	productQuantity
1	Google Sunglasses	20
2	Google Women's Vintage Hero Tee Black	7
3	SPF-15 Slim & Slender Lip Balm	6
4	Google Women's Short Sleeve Hero Tee Red Heather	4
5	YouTube Men's Fleece Hoodie Black	3

Query 08: Calculate cohort map from pageview to addtocart to purchase in last 3 month. For example, 100% pageview then 40% add_to_cart and 10% purchase.

```
with data as(
SELECT
  format_date("%Y%m",parse_date("%Y%m%d", date)) AS month,
  sum(case when hits.eCommerceAction.action_type = "2" then 1 end) as num_product_view,
  sum(case when hits.eCommerceAction.action_type = "3" then 1 end) as num_addtocart,
  sum(case when hits.eCommerceAction.action_type = "6" then 1 end) as num_purchase
FROM
    `bigquery-public-data.google_analytics_sample.ga_sessions_*`,
    UNNEST (hits) hits,
    unnest (hits.product) product
  WHERE _table_suffix BETWEEN "20170101" AND "20170331"
  group by month)

SELECT
  month,
  num_product_view,
  num_addtocart,
  num_purchase,
  (sum(num_addtocart)/SUM(num_product_view)) as add_to_cart_rate,
  (sum(num_purchase)/SUM(num_product_view)) as purchase_rate,

FROM data
group by month, num_product_view, num_addtocart,num_purchase
order by month
```

--Result output:

Row	month	num_product_view	num_addtocard	num_purchase	add_to_cart_rate	add_to_cart_rate_1
1	201701	25787	7342	4328	28.47	16.78
2	201702	21489	7360	4141	34.25	19.27
3	201703	23549	8782	6018	37.29	25.56

MS SQL Sever Querry --- Clean, manipulate data to get the average Month Over Month Percentage of product

```
with CTE_1 as (SELECT [month]

, case when month='Mar' then 3

              when month='Apr' then 4

              when month='May' then 5

              when month='Jun' then 6

               end as month_nbr

      ,[main_category]

         ,SUM(buyer) as Total_buyer

  FROM [SQL Practice].[dbo].[test_1]

  group by month, main_category),


 CTE_2 as (select

month_nbr,

main_category,

Total_buyer,

lag(Total_buyer) over (partition by main_category order by month_nbr) as previous_month,

(lag(Total_buyer) over (partition by main_category order by month_nbr) -
Total_buyer)/lag(Total_buyer) over (partition by main_category order by month_nbr)*100 as
MoM
from CTE_1)

select

main_category,

AVG(MoM) as avg_MoM

from CTE_2

where month_nbr in (4,5,6)

group by main_category

order by AVG(MoM) DESC
```

-- Result Output:

	main_category	avg_MoM
1	Women Clothes	49.8667597213459
2	Moms, Kids & Babies	49.4255431988138
3	Toys	33.6011289510909
4	Voucher & Service	20.4738443793807
5	Computer & Laptop	14.8119831068393
6	Home care	13.5730513603194
7	Watches	13.0878540963121
8	Cameras	12.003822003822
9	Consumer Electronics	10.390154647401
10	Books	-41.6542448779291
11	Sports & Outdoors	-45.0297369621257
12	Others	-47.2592613946203
13	Men Shoes	-47.5220102773817
14	Automotive	-54.7741821888611
15	Home Appliances	-64.6439329364964
16	Mobile & Accessories	-71.9848316685165
17	Women Shoes	-86.9416664887636
18	Kid's Fashion	-121.300285458741
19	Home & Living	-121.422029736233
20	Men Clothes	-123.179633451975
21	Fashion Accessories	-131.240520365223
22	Bags	-151.101742897242
23	Grocery	-211.593089181942
24	Health & Beauty	-298.339160894864